

# SUZAKU Audit Summary

Prepared by Cyfrin

Version 2.0


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# 1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

# 2 About This Document

This document is a public summary of the security audit Cyfrin performed for SUZAKU on SUZAKU. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated July 7, 2025, was delivered to SUZAKU, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

# 3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

# 4 Copyright and License

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# 5 Audit Overview

Over the course of 15 days, the Cyfrin team conducted an audit on the SUZAKU code provided by SUZAKU. The findings consist of 2 Critical, 10 High, 19 Medium & 13 Low severity issues with the remainder being informational and gas optimizations.

**Summary**

| | |
|---|---|
| Project Name | SUZAKU |
| Repository | suzaku-core |
| Commit | 9ce66f5eb093... |
| Audit Timeline | May 20th - Jun 9th, 2025 |
| Methods | Manual Review, Stateful Fuzzing |

**Issues Found**

| Severity | Found | Resolved | Acknowledged |
|---|---|---|---|
| Critical Risk | 2 | 2 | 0 |
| High Risk | 10 | 9 | 1 |
| Medium Risk | 19 | 18 | 1 |
| Low Risk | 13 | 7 | 6 |
| Informational | 1 | 1 | 0 |
| Gas Optimizations | 6 | 6 | 0 |
| Total | 51 | 43 | 8 |

# 6 Audit Scope

The following files were part of the scope for the audit:

```
src/contracts/common/Entity.sol
src/contracts/common/MigratableEntityProxy.sol
src/contracts/common/StaticDelegateCallable.sol
src/contracts/delegator/BaseDelegator.sol
src/contracts/delegator/L1RestakeDelegator.sol
src/contracts/libraries/Checkpoints.sol
src/contracts/libraries/ERC4626Math.sol
src/contracts/middleware/libraries/MapWithTimeData.sol
src/contracts/middleware/libraries/StakeConversion.sol
src/contracts/middleware/AssetClassRegistry.sol
src/contracts/middleware/AvalancheL1Middleware.sol
src/contracts/middleware/MiddlewareVaultManager.sol
src/contracts/rewards/Rewards.sol
src/contracts/rewards/UptimeTracker.sol
src/contracts/service/OperatorL1OptInService.sol
src/contracts/service/OperatorVaultOptInService.sol
src/contracts/vault/VaultTokenized.sol
src/contracts/DelegatorFactory.sol
src/contracts/L1Registry.sol
src/contracts/OperatorRegistry.sol
src/contracts/SlasherFactory.sol
src/contracts/VaultFactory.sol
```

# 7 Risk Classification

| | **Impact: High** | **Impact: Medium** | **Impact: Low** |
|---|---|---|---|
| **Likelihood: High** | Critical | High | Medium |
| **Likelihood: Medium** | High | Medium | Low |
| **Likelihood: Low** | Medium | Low | Low |

# 8 Protocol Summary

Suzaku is a staking infrastructure that enables permissionless validator management and delegation for Avalanche L1 networks. The protocol implements a multi-asset class staking system where operators can run validators backed by delegated stake from multiple vault types, with rewards distribution based on validator uptime and performance.

The key contracts that were part of the audit are:

- **AvalancheL1Middleware:** coordinates operators, validators, and stake accounting across asset classes
- **VaultTokenized:** an ERC-4626 vault for tokenized staking with delegation capabilities
- **L1RestakeDelegator:** manages delegation of vault assets to L1 validators with configurable limits
- **L1Registry:** registry of L1 networks and their middleware configurations
- **OperatorRegistry:** registration system for validator operators
- **VaultFactory and DelegatorFactory:** deployment and lifecycle management of vaults and delegators
- **Rewards:** distributes staking rewards to operators, delegators, and protocol participants
- **UptimeTracker:** monitors validator performance and calculates uptime-based reward eligibility

Assets deposited into the vaults are delegated through the delegator contracts to the validators of registered L1 networks, the middleware accounts for the resulting stake per asset class, and staking rewards are distributed according to validator uptime.

# 9 Attestation

Cyfrin attests that it performed a security review of the SUZAKU code provided by SUZAKU.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-07-07-cyfrin-suzaku-core-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to SUZAKU has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=eaabeffb5a674b89831e9d379f5dd2ca774e474ebc60e5839706c0be08194585
F=2025-07-07-cyfrin-suzaku-core-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.